[INCH-POUND]
A-A-59394
28 September 1999
SUPERSEDING
MIL-H-80003B
28 March 1974

COMMERCIAL ITEM DESCRIPTION

HONING MACHINE, VERTICAL, SINGLE-SPINDLE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-H-80003B for all federal agencies.

1. SCOPE. This commercial item description (CID) covers single-spindle vertical honing machines intended for internal honing of open, blind-end, and interrupted bores.

2. CLASSIFICATION. The honing machines shall be of the sizes shown in table I. The size to be furnished shall be as specified (see 7.3(b)).

3. SALIENT CHARACTERISTICS

3.1 General requirements. The honing machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The honing machine shall, as a minimum, include the following components:

3.2.1 Base and column assembly. The machine base and column assembly shall incorporate materials, shape, and mass to minimize adverse vibration. The base shall have provisions for leveling and anchoring the machine.

3.2.2 Worktable. Unless otherwise specified (see 7.3(c)), the machine shall have either a cast or fabricated, fixed-position, nonadjustable worktable. The worktable surface shall have not less than three T-slots conforming to the requirements of American National Standards Institute (ANSI) B5.1, "T-Slots, Their Bolts, Nuts and Tongues". The worktable shall have a trough around its entire perimeter for collecting spent coolant with means for returning the coolant to the reservoir for reuse.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3419

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TABLE I. Machine sizes.

Size	Maximum bore diameter, inches	Maximum bore length, inches
1	1.5	6
2	1.5	10
3	1.5	20
4	3.5	4
5	3.5	10
6	3.5	20
7	3.5	25
8	3.5	35
9	3.5	40
10	6	10
11	6	20
12	6	36
13	7	40
14	7	180
15	8	18
16	10	10
17	10	20
18	10	30
19	10	40
20	10	50
21	10	65
22	10	180
23	14	180
24	25	180
25	40	175

Note 1: Machine size and capacity shall not be less than the stated requirements selected from table I for the class, swing, and size specified. The ranges of size and capacity shown in table I imply that the machine offered shall be equal to or larger than the size and capacity stated but smaller than the next available size. When the largest machine shown in table I is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent.

3.2.3 Enclosure panels and splash guards. The machine shall be equipped with enclosure panels and splash guards considered standard by the manufacturer.

3.2.4 Spindle. All rotating parts within the spindle head shall be mounted on antifriction bearings. The spindle drive mechanism shall provide infinitely variable or gear-selected spindle speeds as specified (see 7.3(d)).

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3.2.5 Stroke mechanism. The stroke mechanism shall consist of a hydraulic cylinder that reciprocates either the entire spindle head or the spindle quill. The spindle stroke speed shall be infinitely variable over the speed range. The mechanism shall smoothly reciprocate the spindle without bind, vibration, or chatter and shall smoothly accelerate, decelerate, and reverse the direction of spindle travel without shock or lost motion.

3.2.5.1 Stroke reversal control. The stroke mechanism shall be equipped with a means to reverse the spindle stroke at any desired location within the stroke range of the machine. With the machine running at its highest stroke speed and under constant honing load, the spindle stroke shall be repeatable to an accuracy of ±0.03 inch measured at the stroke reversal control setpoints. Machines with a maximum bore diameter of 5 inches or more shall have provisions for short-stroking. When specified (see 7.3(e)), machines with a maximum bore diameter of less than 5 inches shall have a short-stroking mechanism.

3.2.5.2 Spindle dwell. The machine shall have an automatically operated adjustable dwell mechanism that, when activated, will cause the spindle stroke to dwell, or momentarily stop, at the lower end of the stroke for a predetermined adjustable time to facilitate accurate sizing of blind or interrupted bores. When the dwell control is turned to the "off" position, the normal stroke cycle shall resume.

3.2.6 Honing tools. If required, honing tools, honing stones, and other tooling shall be furnished as specified (see 7.3(f)). The honing tools shall be free floating, self-aligning types using universal or ball joints that allow the honing tool to follow the existing bore of the workpiece. The honing tools shall use a positive expansion and contracting system for expanding and contracting replaceable honing stones. The honing tools shall be compatible with the stone expansion mechanism specified in section 3.2.7.

3.2.7 Stone expansion mechanism. The machine shall have an automatic expansion mechanism that shall rapidly expand the honing stones until they are near or in contact with the bore wall, then expand the stones at a slower controlled rate until the desired bore size is reached, and then rapidly collapse the stones while withdrawing the honing tool. The mechanism shall have provisions for manually adjusting the stone expansion rate from the operator control panel and locking the diameter setting after determining the desired size. The stone expansion shall have a positive action mechanism for expanding the stones at a continuous, uniform, and preselected rate. The mechanism shall have a stone wear compensating device that will automatically adjust the stone feed mechanism to compensate for abrasive wear.

3.2.7.1 Spark-out control. If required, the machine shall have a control for stopping expansion of the honing stones while rotation and stroke are continued so that the stones can clean the bore as specified (see 7.3(g)). The control shall allow the time or number of spark-out strokes to be preselected. The control shall be mounted on the operator’s control panel.

3.2.7.2 Stone wear indicator. The machine shall have a stone wear indicator that warns the operator by light, bell, or buzzer when the stones are worn to the extent that workpiece damage is imminent.

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3.2.8 Guide bushing bracket. When specified (see 7.3(h)), the machine shall have a guide bushing bracket that accommodates properly sizes bushings for selected honing bore diameters. Guide brackets on machines with 40 inches or greater spindle travel shall incorporate a power-operated raising and lowering mechanism.

3.2.9 Swarf removal system. Unless otherwise specified (see 7.3(i)), an automatic combination magnetic separator-fabric filter unit shall be furnished.

3.2.10 Control system. The control system shall, as a minimum, provide semiautomatic honing cycle control. The control system shall be capable of completing a honing operation based on preselected settings of spindle speed, stroke speed and length, and dwell for a preselected time or preselected feed rate. Upon completion of the honing cycle, the honing tool shall automatically return to the top of the stroke position and all honing operations shall stop.

3.2.10.1 Operator controls. Unless otherwise specified (see 7.3(j)), the following operator controls and adjustments, as a minimum, shall be provided:

a. Power on/off switch(s)
b. Emergency stop switch
c. Spindle speed
d. Stroke speed
e. Stroke length (upper and lower limits)
f. Dwell
g. Feed rate
h. Coolant on/off switch/valve
i. Cycle start

3.2.10.2 Automatic controls. If required, a programmable logic controller (PLC) shall be provided as specified (see 7.3(k)).

3.2.10.3 Digital readout displays. Unless otherwise specified (see 7.3(l)), the following digital readout displays shall be provided:

a. Spindle speed
b. Stroke speed
c. Stroke setting
d. Cycle time

3.2.11 Coolant system. The coolant system shall be of sufficient capacity to provide a full flow rate of clean coolant during the entire honing process. The system shall be equipped to allow the operator to direct and control the coolant flow to the workpiece.

3.2.12 Coolant refrigeration system. When specified (see 7.3(m)), the coolant system shall be equipped with a self-contained refrigeration system designed to control the coolant temperature between 62° and 68° F throughout the metal removal range of the honing machine.

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3.2.13 Hydraulic system. When specified (see 7.3(n)), the honing machine shall have a hydraulic system to operate the spindle stroke and the hone feed mechanism. The hydraulic system shall have overpressure protection to prevent damage to components. A filter system shall ensure delivery of clean fluid to the hydraulic system. The hydraulic system reservoir shall have a sight gauge to indicate fluid level. The hydraulic fluid temperature shall be maintained within the proper operating range during machine operation. A means shall be provided to drain and clean the hydraulic reservoir.

3.2.14 Electrical system. The electrical system shall conform to the National Fire Protection Association (NFPA) 79, "Machinery, Industrial, Electrical Standard For". The machine shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit and shall be wired for the voltage specified (see 7.3(o)). The control system shall operate on a 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.15 Electric motors. The machine drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.3 (p)), the motors shall have dripproof enclosures.

3.2.16 Optional accessories. Optional accessories shall be furnished as specified (see 7.3(q)).

3.3 Performance. Unless otherwise specified (see 7.3(r)), the machine shall meet the alignment requirements shown in table II and the performance requirements shown in table III when tested in accordance with section 5.3 herein.

TABLE II. Alignment requirements.

Parameter	Tolerances
Spindle:	
Runout, measured at spindle nose	0.001 inch, total indicator readout
End play	0.0002 inch
Spindle axis square to worktable surface	0.001 inch per foot
Spindle head travel square to worktable surface	0.001 inch per foot
Worktable:	
Working surface flat in longitudinal and transverse direction	0.001 inch per foot

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TABLE III. Performance requirements.

Machine size	Bore parameters			
	Maximum out-of-round[1], inches	Maximum straightness deviation, inches	Surface finish[2], microinches, R_a	Minimum metal removal rate, cubic inches per minute
1	0.0001	0.0002	5	0.05
2	0.0001	0.0002	5	0.05
3	0.0001	0.0002	5	0.05
4	0.0002	0.0002	5	0.50
5	0.0002	0.0002	5	0.50
6	0.0002	0.0002	5	0.50
7	0.0002	0.0002	5	0.50
8	0.0002	0.0002	5	0.50
9	0.0002	0.0002	5	0.50
10	0.0003	0.0004	5	0.75
11	0.0003	0.0004	5	0.75
12	0.0003	0.0004	5	0.75
13	0.0003	0.0004	5	1.00
14	0.0003	0.0004	5	1.00
15	0.0003	0.0004	5	1.50
16	0.0005	0.0004	5	2.00
17	0.0005	0.0004	5	2.00
18	0.0005	0.0004	5	2.00
19	0.0005	0.0004	5	2.00
20	0.0005	0.0004	5	2.00
21	0.0005	0.0004	5	2.00
22	0.0005	0.0004	5	2.00
23	0.0005	0.0005	5	2.50
24	0.0005	0.0005	5	2.50
25	0.0005	0.0005	5	2.50

[1] Measured at random intervals throughout bore length.
[2] Surface finish equal to or better than the value indicated. Surface finish, R_a, is the arithmetic average of the distance of the roughness profile from the mean line in accordance with ANSI B46.1.

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3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions as specified (see 7.3(s)).

3.5 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with ANSI B11.9, "Machines, Grinding, Safety Requirements For The Construction, Care, And Use Of", and Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.3(t)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.3(u)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.3(v)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer’s name
c. Machine model designation
d. Power input (volts, total amperes, phase, frequency)
e. Short-circuit/over-current rating
f. Contract number or order number
g. National stock number
h. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.3(w)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The honing machine shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user’s health and shall be

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adhered to during the manufacturing, service, transportation, storage, and operation/use of the machine.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The honing machine shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same machine offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Inspection. The machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.3(x)), the machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Alignment accuracy test. The machine shall be tested to determine compliance with the alignment accuracy requirements shown in table II.

5.3.2 Operational test. The machine shall be operated in accordance with the manufacturer's standard operating procedures. The machine shall be operated at no load for not less than 60 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The machine shall be subjected to the tests in sections 5.3.3.1 through 5.3.3.3. The choice of stones, coolant, stone feed rate, rotating and stroking speeds, and fixturing shall be the option of the supplier.

5.3.3.1 Test workpiece. Machines having a boring capacity less than 16 inches in diameter shall be tested using a section of American Iron and Steel Institute/Society of Automotive Engineers (AISI/SAE) 1026 cold-drawn seamless steel tubing. Machines having a boring capacity over 16 inches in diameter shall be tested using a workpiece fabricated from rolled and welded steel plate. The work piece shall have an inside diameter not less than 75 percent of the machine's maximum bore diameter and a length not less than 75 percent of the machine's maximum bore length. No pretest preparation of the workpiece shall be made to correct any inaccuracies that may be inherent to the workpiece as received from the mill.

5.3.3.2 Honing accuracy test. The test shall be performed using only one rough honing operation and one finish honing operation. The finished bore dimensions shall conform to the roundness, straightness, and surface finish requirements shown in table III.

5.3.3.3 Metal removal rate test. The metal removal rate shall be demonstrated on the workpiece used in the honing accuracy test in section 5.3.3.2. The metal removal rate shall conform to the requirements shown in table III.

5.4 Acceptance. Unless otherwise specified (see 7.3(y)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment accuracy tests shall be

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compared with the requirements shown in table II. The results of the performance tests shall be compared with the performance requirements listed shown in table III. Failure of the machine to meet the requirements shown in tables II and III shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.3(z)).

7. NOTES

7.1 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

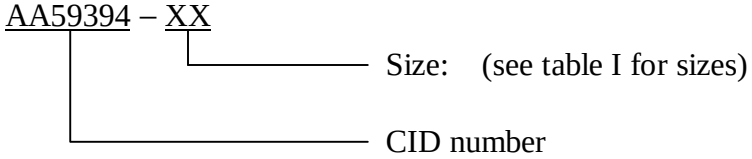
7.1.3 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 OSHA standards. Copes of OSHA standards may be obtained from the U.S. Department of Labor, 300 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.1.6 AISI standards. Copies of AISI/SAE standards may be obtained from the American Iron and Steel Institute, 1101 17th Street NW, Suite 1300, Washington, DC 20036.

7.2 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



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7.3 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Size of honing machine required (see 2)
c. Worktable, if different (see 3.2.2)
d. Spindle speeds, infinitely variable or gear-selected, as specified (see 3.2.4)
e. Short-stroking mechanism, if required (see 3.2.5.1)
f. Honing tools, stones, and other tooling, as specified (see 3.2.6)
g. Spark-out control, as specified (see 3.2.7.1)
h. Guide bushing bracket, if required (see 3.2.8)
i. Automatic combination magnetic separator-fabric filter unit, if different (see 3.2.9)
j. Operator controls, if different (see 3.2.10.1)
k. Programmable logic controller, if required (see 3.2.10.2)
l. Digital readout displays, if different (see 3.2.10.3)
m. Coolant refrigeration system, if required (see 3.2.12)
n. Hydraulic stroke and feed mechanism, if required (see 3.2.13)
o. Voltage requirements, specify (3.2.14)
p. Dripproof enclosures, if required (see 3.2.15)
q. Optional equipment, as specified (see 3.2.16)
r. Performance requirements, if different (see 3.3)
s. Machine dimensions (length, width, and height) and weight, as specified (see 3.4)
t. Measuring and indicating device units of measure, as specified (see 3.6)
u. Dual graduation of indicating devices, if required (see 3.6)
v. Nameplate, if different (see 3.7)
w. Lubrication plate or chart, if required (see 3.8)
x. Testing, if different (see 5.3)
y. Acceptance tests site, if different (see 5.4)
z. Packaging requirements, as specified (see 6.1)

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Navy - MC

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3419-0231)

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